Software Requirements Specification

[Project Name]

Aligned with IEEE 830-1998 and ISO/IEC/IEEE 29148:2018

Author: Athulya Peshali, Business Analyst

Document ID: SRS-*[PROJ]*-001 **Version:** 0.1

Date: *[YYYY-MM-DD]*

© 2026 Athulya Peshali. All rights reserved.

Authored by Athulya Peshali (Business Analyst). Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

This template contains no confidential client or employer information. All project names, data and requirements shown are fictional examples.

© 2026 Athulya Peshali. All rights reserved. Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

Document Control

Document Title	Software Requirements Specification: *[Project Name]*
Document ID	SRS-*[PROJ]*-001
Version	0.1 (Draft)
Status	Draft / In Review / Approved / Baselined
Author	Athulya Peshali, Business Analyst
Reviewed By	*[Name, Role]*
Approved By	*[Name, Role]*
Date Created	*[YYYY-MM-DD]*
Last Updated	*[YYYY-MM-DD]*
Classification	Portfolio Sample – Not for Redistribution

Revision History

Version	Date	Author	Description of Change
0.1	*[YYYY-MM-DD]*	Athulya Peshali	Initial draft

Approval Sign-off

Name	Role	Signature	Date
	Project Sponsor		
	Project Manager		
	Technical Lead		
	QA Lead		

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

1. Introduction

1.1 Purpose

State the purpose of this SRS and its intended audience.

Example: This document specifies the functional and non-functional requirements for *[Project Name]*, version *[x.x]*. It is intended for the development team, QA, project manager and business stakeholders.

1.2 Scope

Name the product, say what it will and will not do, and describe its benefits and goals.

- **Product name:** *[Name]*
- **In scope:** *[Key capabilities]*
- **Out of scope:** *[Explicit exclusions]*
- **Business objectives:** *[Measurable goals, e.g. reduce processing time by 30%]*

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
BA	Business Analyst
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 References

ID	Document	Version	Source
REF-01	IEEE Std 830-1998	1998	IEEE
REF-02	ISO/IEC/IEEE 29148:2018	2018	ISO
REF-03	*[Business Requirements Document]*		

1.5 Document Overview

Briefly describe how the rest of the document is organised.

2. Overall Description

2.1 Product Perspective

How does this product fit into the wider system or business context? Is it new, a replacement, or part of a larger system?

> *[Insert context diagram here: Users → Your System → Database / External Systems A and B]*

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

2.2 Product Functions (High-Level)

- *[Function 1: e.g. user registration and authentication]*
- *[Function 2: e.g. order management]*
- *[Function 3: e.g. reporting dashboard]*

2.3 User Classes and Characteristics

User Class	Description	Technical Skill	Frequency	Priority
Administrator		High	Daily	High
Standard User		Medium	Weekly	High
Guest		Low	Occasional	Low

2.4 Operating Environment

- **Client:** *[Browsers, OS, devices]*
- **Server:** *[Hosting, OS, runtime]*
- **Database:** *[DBMS]*
- **Integrations:** *[Third-party systems]*

2.5 Design and Implementation Constraints

- *[Regulatory, e.g. PDPA (Sri Lanka), GDPR]*
- *[Technology mandates, budget, timeline, standards]*

2.6 Assumptions and Dependencies

ID	Type	Description
A-01	Assumption	*[e.g. Users have stable internet access]*
D-01	Dependency	*[e.g. Payment gateway API availability]*

3. Stakeholder and User Requirements

ISO 29148 asks for stakeholder needs to be captured and linked to system requirements.

3.1 Stakeholder Register

ID	Stakeholder	Role	Interest / Need	Influence
STK-01		Sponsor		High
STK-02		End User		Medium

3.2 Business Requirements

ID	Requirement	Rationale	Priority
BR-01	*[The business needs to...]*	*[Why]*	Must

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

3.3 User Stories (Optional)

ID	As a...	I want to...	So that...	Acceptance Criteria
US-01	*[role]*	*[goal]*	*[benefit]*	*[Given / When / Then]*

4. Specific Requirements

> **Requirement writing rules (ISO 29148):** each requirement should be necessary, unambiguous, complete, singular, feasible, verifiable and consistent.
>
> Use **"shall"** for mandatory requirements and **"should"** for desirable ones. Avoid vague terms such as fast, user-friendly, etc., as appropriate.
>
> **Priority (MoSCoW):** Must / Should / Could / Won't

4.1 External Interface Requirements

4.1.1 User Interfaces

ID	Requirement	Priority	Source
UI-01	The system shall provide a responsive interface that works on screen widths from 360 px to 1920 px.	Must	STK-02

4.1.2 Hardware Interfaces

ID	Requirement	Priority	Source
HI-01			

4.1.3 Software Interfaces

ID	Requirement	Priority	Source
SI-01	The system shall integrate with *[External API]* over HTTPS using REST/JSON.	Must	

4.1.4 Communication Interfaces

ID	Requirement	Priority	Source
CI-01	All client-server communication shall use TLS 1.2 or higher.	Must	

4.2 Functional Requirements

4.2.1 [Feature / Module Name, e.g. User Management]

ID	Requirement	Priority	Source	Acceptance Criteria
FR-001	The system shall allow a new user to register using a valid email address and password.	Must	US-01	Registration succeeds with valid input; an error is shown for invalid input.
FR-002	The system shall send a verification email within 60 seconds of registration.	Should		

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

ID	Requirement	Priority	Source	Acceptance Criteria
FR-003				

4.2.2 [Feature / Module Name]

ID	Requirement	Priority	Source	Acceptance Criteria
FR-010				

4.3 Use Case Specification (Sample)

Use Case ID	UC-01
Name	*[e.g. Place Order]*
Primary Actor	*[Customer]*
Preconditions	*[User is logged in]*
Trigger	*[User selects "Checkout"]*
Main Flow	1. *[Step]* 2. *[Step]* 3. *[Step]*
Alternative Flows	*[A1: Payment fails → show error and allow retry]*
Postconditions	*[Order is saved with status "Pending"]*
Related Requirements	FR-001, FR-010

4.4 Non-Functional Requirements

4.4.1 Performance

ID	Requirement	Priority
NFR-P-01	The system shall respond to 95% of page requests within 2 seconds under a load of 500 concurrent users.	Must

4.4.2 Security

ID	Requirement	Priority
NFR-S-01	Passwords shall be stored using a salted, adaptive hash (e.g. bcrypt or Argon2).	Must
NFR-S-02	The system shall enforce role-based access control for all protected resources.	Must

4.4.3 Reliability and Availability

ID	Requirement	Priority
NFR-R-01	The system shall achieve 99.5% monthly uptime, excluding planned maintenance.	Should

4.4.4 Usability and Accessibility

ID	Requirement	Priority
NFR-U-01	The user interface shall conform to WCAG 2.1 Level AA.	Should

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

4.4.5 Maintainability and Portability

ID	Requirement	Priority
NFR-M-01		

4.4.6 Legal, Regulatory and Compliance

ID	Requirement	Priority
NFR-L-01	The system shall comply with applicable data-protection law (e.g. Sri Lanka PDPA No. 9 of 2022).	Must

4.5 Data Requirements

Entity	Key Attributes	Retention	Notes
User	id, name, email, role	*[x years]*	

4.6 Other Requirements

Installation, localisation, documentation, training and similar needs.

5. Verification and Validation

ISO 29148 expects each requirement to have a stated verification method.

Method	Description
Inspection	Review of documents or code
Analysis	Calculation, modelling or simulation
Demonstration	Operating the system to show behaviour
Test	Executing test cases against measurable criteria

Req. ID	Verification Method	Test Case ID	Status
FR-001	Test	TC-001	Not started
NFR-P-01	Test (load)	TC-050	Not started

6. Requirements Traceability Matrix (RTM)

Business Req.	Stakeholder	User Story	Functional Req.	Test Case	Status
BR-01	STK-01	US-01	FR-001, FR-002	TC-001	Draft

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.

7. Appendices

Appendix A: Requirements Change Log

CR ID	Date	Requested By	Description	Impact	Decision

Appendix B: Open Issues and TBDs

ID	Issue	Owner	Due Date	Status
TBD-01				Open

Appendix C: Supporting Models

Add BPMN diagrams, ER diagrams, wireframes or state diagrams here.

Prepared for portfolio verification. Not for unauthorised distribution or reproduction.